Ling 285
Human Language and Technology
Lecture 4+5

Dynamical Systems 3

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HW Due Friday at 5pm

Harmonic Oscillator: Elements x, v

- We are interested in how x and v will change in time, *due to the their relation*.
- We will use the symbol $\dot{x}$ to mean the change of x as time increases. $\dot{v}$ is the change in v as time increases.
- $\dot{x}$ is positive means x rises.
- $\dot{x}$ is negative means x falls
- Same for v.
- This is Isaac Newton's notation.
- Harmonic Oscillator: x and v change each other!

$$\dot{x} = v$$

$$\dot{v} = -x$$

Harmonic Oscillator: Elements x, v Example

- We are interested in how x and v will change in time, *due to the their relation*.
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- This is Isaac Newton's notation.
- Harmonic Oscillator: x and v change each other!

v is -5. Therefore, x will go down!

$$\dot{x} = v$$

$$\dot{v} = -x$$

x is -5. Therefore, v will go UP!

Harmonic Oscillator: Elements x, v

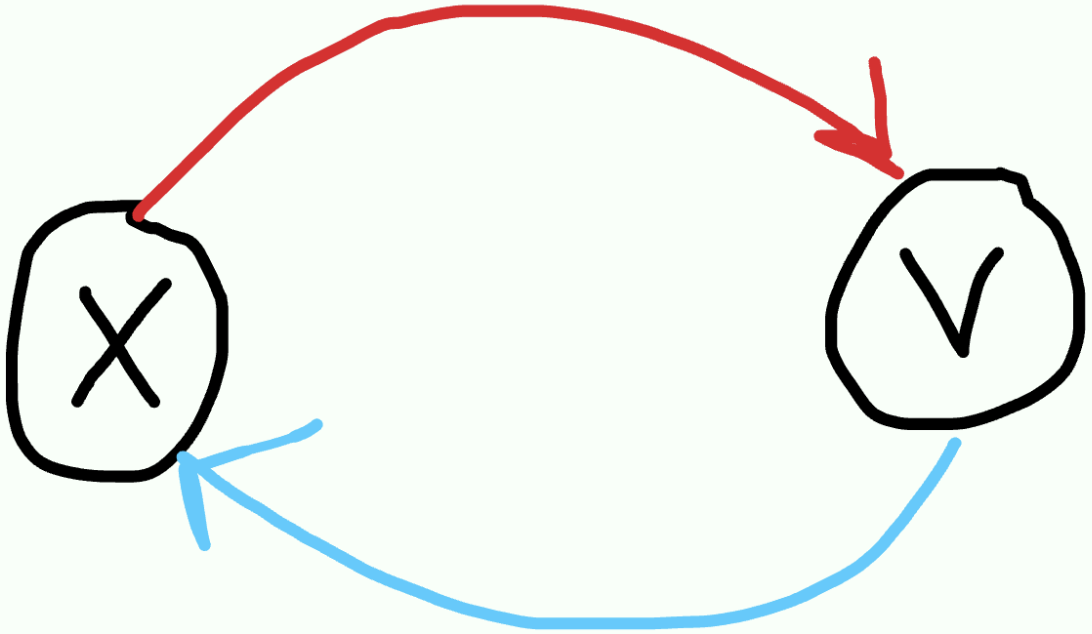
- Meaning of $\dot{x} = v$:
  - v positive: $\dot{x}$ positive, so x rises
  - v negative: $\dot{x}$ negative, so x falls
- Meaning of $\dot{v} = -x$:
  - x positive: $\dot{v}$ negative, so x falls
  - x negative: $\dot{v}$ positive, so x rises

$$\dot{x} = v$$

$$\dot{v} = -x$$

x-v System

Oscillation/Vibration Dynamical System



HW Due Friday at 5pm

Everything’s a harmonic oscillator, if you’re brave enough

Slice of air in your vocal tract

High pressure means air very close

Low pressure means air very far apart

There’s a cosine in your mouth!

Vowels

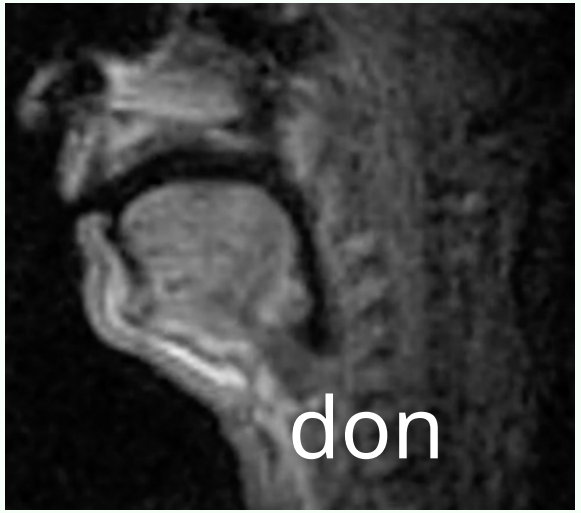
High





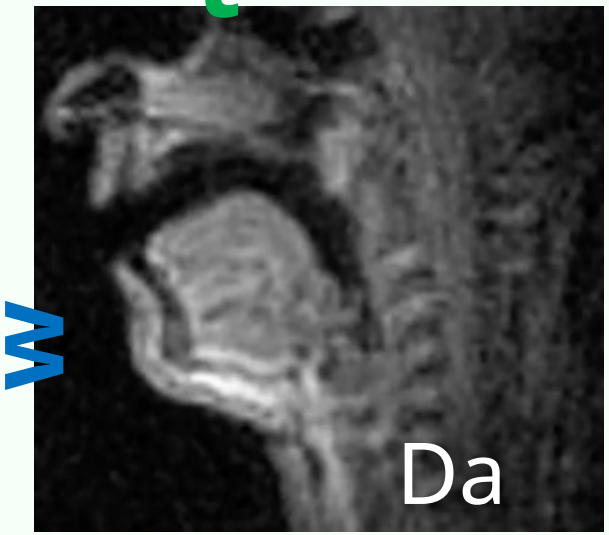
Front

Back



Central

Low



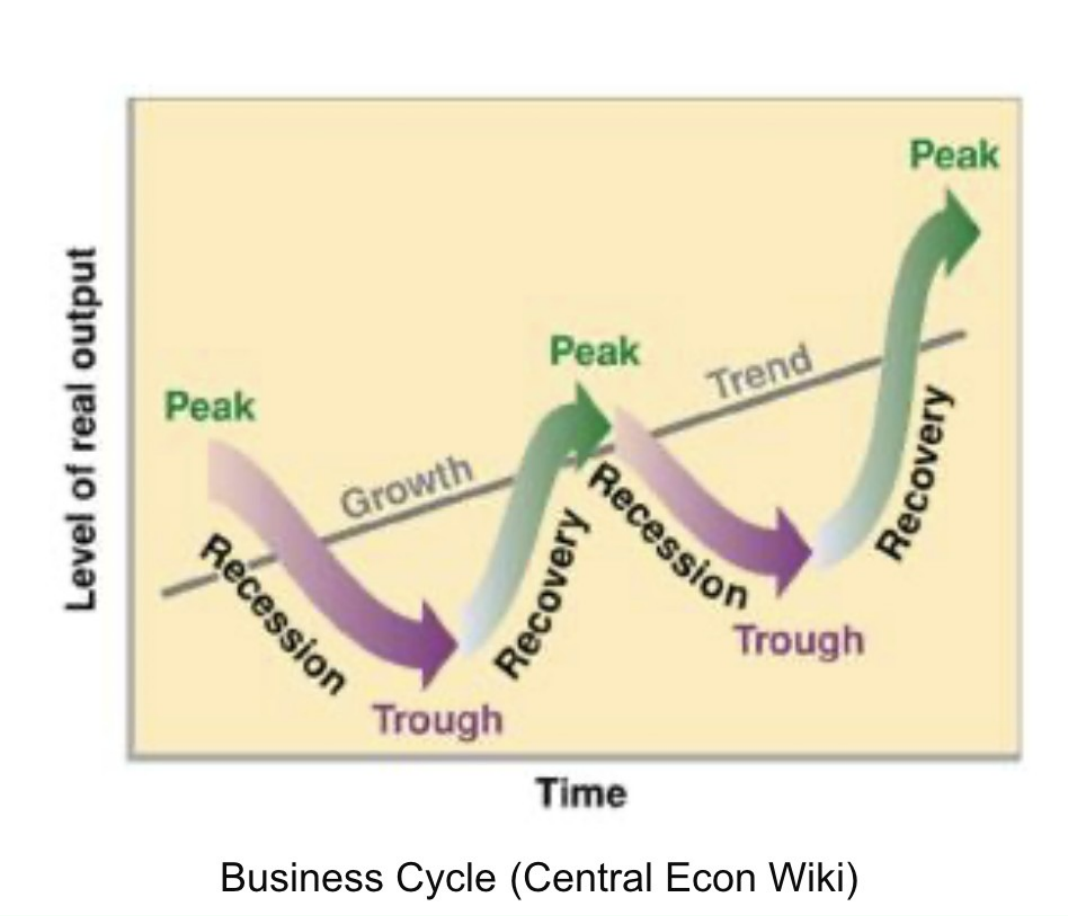


Praat Demonstartion

2 things are vibrating. The air in your vocal tract, as above, but also your vocal folds

- https://www.youtube.com/watch?v=9Tlpkdq8a8c
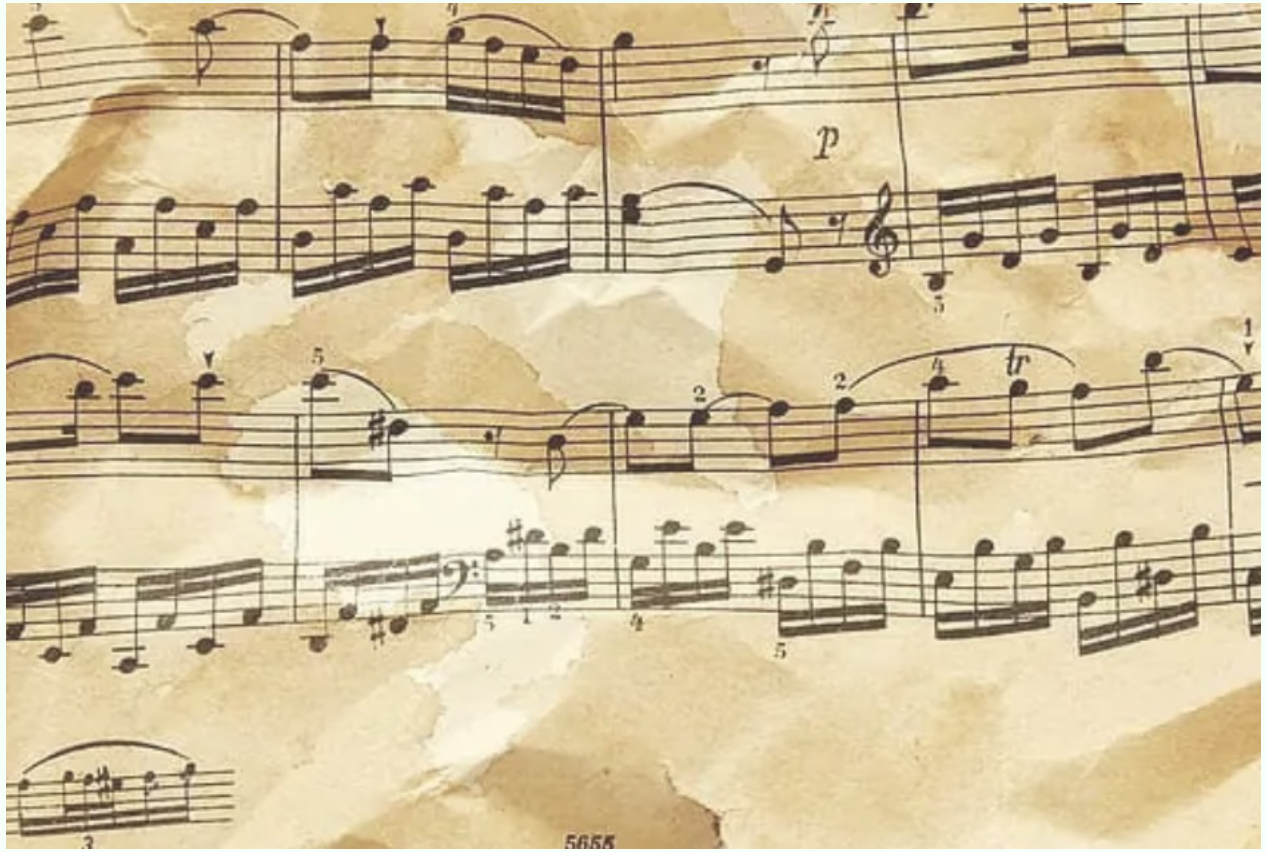
Harmonic Oscillator in Economics



Business Cycle (Central Econ Wiki)

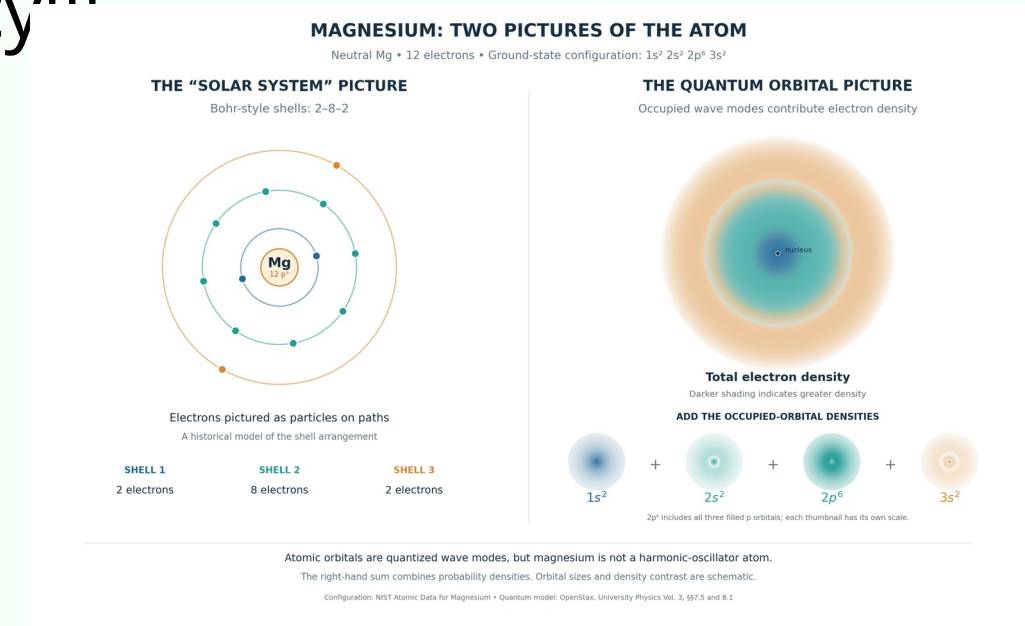
New Reading from Meadows online!

Music is all about Harmonic oscillators, of course



Who is made of matter?

- https://en.wikipedia.org/wiki/Quantum_harmonic_oscillator
- Matter is made up of atoms with different numbers of electrons, some vibrating at a low frequency and some at a high frequency!!!



x-v System

Let’s do this in Python!!!